

Alejandro Zárate Macías

AI Engineer | Generative AI, RAG & Agentic Systems

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PROFESSIONAL SUMMARY

AI Engineer with nearly 5 years of software engineering experience and 2+ years delivering production ML and GenAI systems across public-sector and industrial environments. Builds end-to-end RAG, agentic, and ML solutions, from retrieval and FastAPI services to PostgreSQL data layers, cloud deployment, monitoring, and user-facing integrations. Hands-on with Python, FastAPI, LangChain, FastMCP, Airflow, PostgreSQL/PostGIS, Docker, Linux, GCP, and AWS. M.Sc. candidate researching Spanish-optimized embeddings for Agentic-LLM tool selection and context retrieval.

CORE SKILLS

Generative AI: RAG, vector search, embeddings, semantic search, agentic workflows, tool/function calling, MCP/FastMCP, Deep Agents, LangChain, prompt engineering

Backend & APIs: Python, FastAPI, SQLAlchemy, REST APIs, async services, PostgreSQL, PostGIS, SQL

Machine Learning & MLOps: PyTorch, NLP, OCR, computer vision, CNNs, ONNX Runtime, MLflow, DVC, model evaluation, experiment tracking

Data & Infrastructure: Apache Airflow, ETL/ELT, DuckDB, Docker, Linux, GCP, AWS, Terraform, CI/CD, GitHub Actions

AI-Assisted Engineering: Claude Code, Codex, OpenCode, GitHub Copilot, reusable skills, coding agents, AI-assisted code review

PROFESSIONAL EXPERIENCE

Institute of Statistical and Geographic Information of Jalisco (IIEG) | Guadalajara, Mexico

AI Developer | Jan 2026 – Present

- Build and deploy GenAI applications that let institutional users retrieve, analyze, and interact with structured and unstructured government data.
- Design production RAG pipelines spanning document ingestion, vector retrieval, context ranking, PostgreSQL data, and institutional API integration.
- Develop agentic workflows with planning, subagents, tool calling, document reading, sandboxed execution, reusable skills, and MCP integrations.
- Build FastMCP servers and Python/FastAPI services that expose internal APIs, data services, and ML inference workflows to AI assistants.
- Integrate AI services with Airflow ETL pipelines, PostgreSQL/PostGIS, Linux servers, Docker, and cloud infrastructure.

Software Project Leader | Jun 2025 – Jan 2026

- Led delivery of internal software and LLM information-retrieval systems for government data products.
- Coordinated ETL development and production operations across Linux and cloud environments.

Systems Specialist | Dec 2024 – Jun 2025

- Developed web-based data collection systems for government institutions and contributed to RAG-enabled internal assistants.
- Built and maintained Apache Airflow ETL pipelines and Apache Superset analytics platforms.
- Implemented object-detection and OCR models for road-infrastructure analysis.

Intelica

Software and Artificial Intelligence Developer (Consulting Project) | Dec 2025 – Mar 2026

- Developed a platform for entering and managing product-demand forecast values supported by forecasting models.
- Deployed the platform on AWS using Terraform, Amazon Aurora, AWS Lambda, and serverless frontend hosting.
- Configured shared AI coding agents that helped the development team generate standardized, consistent code.

Machine Learning Engineer (Consulting Project) | Mar 2024 – Jul 2024

- Developed production inference APIs with FastAPI and ONNX Runtime.
- Implemented MLOps workflows with MLflow and DVC, including experiment and artifact management.
- Automated training and deployment with GitHub Actions and built Label Studio data-labeling pipelines.

Jalisco State Government

Machine Learning Engineer | Sep 2023 – Dec 2024

- Built AI data-processing pipelines and production REST APIs for model inference.
- Designed repeatable training, evaluation, and experimentation workflows for ML models.
- Implemented CNN models for image classification and segmentation.
- Installed and maintained NVIDIA DGX A100 infrastructure for deep-learning workloads.

Bosch

Embedded Software Developer Jr | Aug 2022 – Aug 2023

- Supported embedded software for automotive production projects and resolved production code issues.
- Managed software releases and customer deliveries.

Software Automation Intern | Aug 2021 – Aug 2022

- Built scripts and automation tools for testing teams while learning industrial requirements, Scrum, Kanban, and cross-functional software delivery.

SELECTED AI SYSTEMS

- **Institutional RAG Assistant:** Enterprise assistant over institutional documents and databases using retrieval pipelines, vector search, FastAPI, PostgreSQL, and internal APIs.
- **Agentic AI Tooling Platform:** Multi-agent workflows with planning, subagents, document reading, sandboxed execution, reusable skills, and MCP integrations.
- **MCP Integration Layer:** FastMCP servers that expose internal APIs and data services through standardized AI-consumable tools.

EDUCATION

University of Guadalajara | M.Sc. in Data Science Engineering, in progress | Expected Aug 2027

Thesis: Spanish-optimized embeddings for Agentic-LLM tool filtering and context retrieval.

University of Guadalajara | B.Sc. in Computer Science Engineering | Aug 2018 – Dec 2022

Graduated with academic excellence.

LANGUAGES

Spanish - Native | English - B2, professional technical communication